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[5668]-201

S.E. (Information Technology) (First Semester) EXAMINATION, 2019

DISCRETE STRUCTURES

(2015 PATTERN)

Time : Two Hours

Maximum Marks : 50

N.B. :— (i) Solve Q. Nos. 1 or 2, 3 or 4, 5 or 6, 7 or 8.

(ii) Neat diagrams must be drawn wherever necessary.

(iii) Figures to the right indicate full marks.

(iv) Assume suitable data, if necessary.

1. (a) A bag contains 3 red, 6 white and 7 blue balls. What is the probability that two balls drawn are white and blue ? [6]

(b) Three cards are drawn from a well-shuffled pack of 52 cards. Find the probability that they are a king, a queen and a jack. [6]

Or

2. (a) How many 3-digit numbers can be formed from the digits 2, 3, 5, 6, 7 and 9, which are divisible by 5 and none of the digits is repeated ? [6]

(b) What is Multiset ? Let A and B be the multisets $\{a, a, b, b, c, f\}$ and $\{a, a, b, b, b, d, d\}$, respectively. [6]

Find :

(a) $A \cup B$

(b) $A \cap B$

(c) $A - B$

(d) $B - A$.

3. (a) Prove by Mathematical Induction that for $n \geq 1$: [6]

$$1.1! + 2.2! + 3.3! + \dots + n.n! = (n+1)! - 1.$$

P.T.O.

(b) Define with example : [6]

(i) Equivalence relation

(ii) POSET

(iii) Lattice.

Or

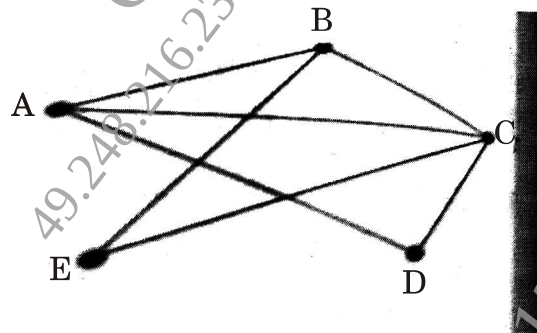
4. (a) Solve the following recurrence relation : [6]

$$x(n) - 6x(n-1) + 9x(n-2) = 0$$

$$x(0) = 0$$

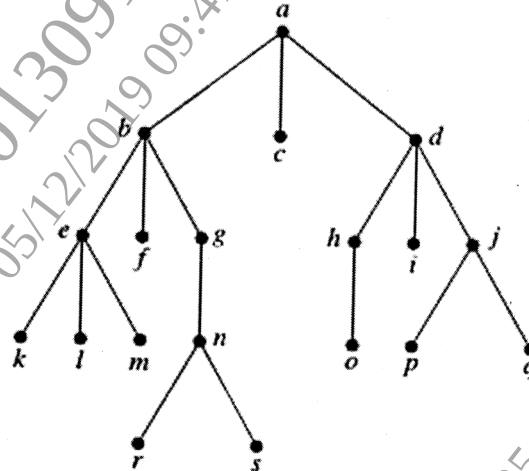
$$x(1) = 3.$$

(b) Consider the graph given in the figure, find the set $V(G)$ of the vertices's present in G and that set $E(G)$ of edges of G also find the degree of each vertex and show that sum of the degree of the vertices's is twice the number of edges in graph G : [6]



5. (a) Build a binary search tree for the words *banana*, *peach*, *apple*, *pear*, *coconut*, *mango*, and *papaya* using alphabetical order. Write sequence of visiting words in preorder and post-order traversal. [7]

- (b) Determine the order in which a preorder, postorder and inorder traversal visits the vertices of the given ordered rooted tree. [6]



Or

6. (a) What is expression tree ?

Represent the expressions

(i) $(x + xy) + (x/y)$

(ii) $x + ((xy + x)/y)$

[7]

using binary trees. Write each of these expressions in

(a) prefix notation.

(b) postfix notation.

- (b) For the following set of characters, construct Huffman code.

Find average bit length of the code :

[6]

Character	A	B	C	D	E
Frequency	0.1	0.15	0.25	0.2	0.3

7. (a) Let $G = \{\text{even, odd}\}$ and binary operation $\oplus$ be define as : [7]

$\oplus$	even	odd
even	even	odd
odd	odd	even

Show that $(G, \oplus)$ is a group.

- (b) Define the following : [6]

- (a) Group
(b) Monoid
(c) Abelian group.

Or

8. (a) Show that $(G, +_6)$ is an abelian group where $G = \{0, 1, 2, 3, 4, 5, 6, 7\}$. [7]
- (b) Prove that $G = \{0, 1, 2, 3, 4, 5\} = \mathbb{Z}_6$ is an abelian group of order 6 with respect to addition modulo 6. [6]